

Scientific workflows span specialized programs, remote schedulers, and local analysis tools. Job completion alone does not prove that scientifically valid outputs were produced or that the correct files reached the next stage.

Workflow Portal and Scientific Verification

KEY CONTRIBUTION

We transformed ePolyScat from a collection of disconnected tools into a guided, inspectable workflow. Each stage preserves the researcher's original input, records output provenance, and exposes a verified handoff to the next calculation or visualization step.

Evidence narrative: from input to interpretation

01 Configure

New Run

Modules: Single application run

Utilities: Post-processing utility

Workflows: Trained calculation

1 Data Generation 2 ePolyScat Run 3 Post-processing 4 Visualization

Input Computation

Gaussian16: Input computation

OpenMolcas: Input computation

Required Files

Molcas_Input: Molcas input file

Missing

Descriptions (H406)

WORKFLOW: Pre-constructed sequences of modules and utilities designed to carry out entire processes

Data_Gen: Data_Generation

OpenMolcas: Computes initial wavefunctions for the ePolyScat Calculations

Input Files

Molcas_Input

02 Preserve

Data Entry

Grid / Expansion State Definitions Potentials Energies / Partial Waves Outputs **Ordered Sequence**

Ordered Sequence

Data records Commands LMax

1 LMax LMax: 15

2 LMaxi LMaxi: 120

3 EMax EMax: 50

4 Feigenfng Feigenfng: 13

5 Scatfng Scatfng: 0.0001 0.01 0.5

03 Verify

ePolyScat Run on 2026/7/19 09:14:18

Modules: EPOLYSCAT.DMAT

CONTINUE TO Visualization & Analysis

Input

epolyscat_chd_test 01: input_data.mold en

Output

epolyscat_chd_test 01: inp

epolyscat_stderr ePolyScat.stdout

sp_input test01Bfms.dat

job_1316236077.drum ePolyScat.out

AT221151778

04 Interpret

Interpret

new1clean.molden
Model 1.1 MD 792 alpha
Energy = -8.6161
Symmetry = 1.2
Occupancy = 2

test01clean.molden
Model 1.1 MD 792 alpha
Energy = -8.6161
Symmetry = 1.2
Occupancy = 2

Continue verified results into post-processing and local scientific visualization.

Engineering decisions

- Lossless ordered document model for reproducible ePolyScat inputs.
- Semantic output roles instead of filename-only inference.
- Resource-aware execution contracts across supported HPC systems.
- Provenance-aware continuation with researcher review before submission.

Scientific value

Researchers can reproduce exactly what was submitted, distinguish scheduler artifacts from scientific results, and continue a calculation without manually rebuilding file dependencies.

Reproducible input → interpretable output

ACCEPTANCE RESULT

Portal configuration → HPC execution → scientific proof → provenance-aware handoff → local visualization is operational and demonstrated with real runs

Acceptance evidence

296 Automated Checks Passed

3 HPC Platforms Validated

2 Scientifically Verified Runs

4 Operational Workflow Stages

Non-gRPC test coverage, HPC wrapper regressions, and the production build passed successfully.

ACCEPTANCE CLAIM

Connect configuration, execution, scientific validation, and stage-to-stage handoff in one portal experience.

[Partner Progress Title]

[One sentence: what changed, for whom, and why it matters.]

Problem & method

[Problem and method: summarize the approach in two or three concise points.]

Implementation

[Implementation or experimental contribution: name the system, artifact, or analysis delivered.]

Primary result

PRIMARY RESULT FIGURE

Replace `\PartnerVisual` with
`\includegraphics[width=\linewidth]{...}`

[One sentence explaining what the visual proves.]

Validation evidence

✓ [Tests, measurements, comparison, or user evidence supporting the claim.]

Outcome metrics

[Two to four defensible outcome metrics, separated by centered dots.]

How this contribution fits the shared objective

The diagram illustrates the three steps of the research process, each in a colored box with a number and a description:

- 01** Traceable configuration (Blue box)
- 02** Verified computation (Teal box)
- 03** Reviewable research output (Orange box)

REPLACEMENT GUIDE

Edit the eight \Partner... macros near the top; every labeled area updates automatically